Question #1 of 59

Question ID: 1473823

The Austrian private equity firm RD primarily makes leveraged buyout investments as the firm's management strongly believes that debt makes companies more efficient. The *least likely* explanation of management's rationale is to:

A) transfer risk.

B) increase firm efficiency.

C) reduce the interest tax shield.

Explanation

A PE firm's debt is frequently securitized and repackaged as collateralized debt or loan obligations, resulting in a transfer of risk to the debt buyer. Greater use of debt also requires disciplined and timely payment of interest, causing a PE firm's portfolio companies to use free cash flow efficiently. Higher leverage generally increases the tax savings from the use of debt (the interest tax shield) increasing firm value in the meantime.

(Module 33.1, LOS 33.a)

Question #2 of 59

Question ID: 1473853

The party in a private equity fund that has unlimited liability for the firm's debts, and this party's share in fund profits, respectively, is referred to as:

	Unlimited liability	Share in fund profits
A)	Limited partner	Distribution waterfall
B)	General partner	Carried interest
C)	Manager	Management fees

Explanation

Limited partners' liability does not extend beyond their capital investment, whereas general partners (the fund managers) have unlimited liability for the firm's debt. The general partner's share in fund profits is referred to as *carried interest*. Management fees are paid annually as a percentage of capital (NAV, paid-in-capital, or committed capital) and are not tied to fund profits.

(Module 33.2, LOS 33.g)

Question #3 of 59

Question ID: 1473839

A private equity firm is guaranteed to receive 80% of the residual value of a leveraged buyout investment, with the remaining 20% owing to management. The initial investment is $500 million, and the deal is financed with 70% debt and 30% equity. The projected multiple is 2.0. The equity component consists of:

- $120 million preference shares.
- $25 million private equity firm equity.
- $5 million management equity.

At exit in 5 years the value of debt is $150 million and the value of preference shares is $300 million. The payoff multiple for the private equity firm and for management, respectively, is *closest* to:

Private equity ; Management

A)	5.10	22.0
B)	3.03	11.0
C)	6.34	46.0

Explanation

The calculations at exit are as follows (all in million $):

- The exit value will be $500 × 2.0 (the specified multiple) = $1,000.
- Outstanding debt is $150.
- Preference shares are worth $300.

- Private equity firm's value: 80% of the residual exit value:

  (0.80)($1,000 − $150 − $300) = $440.

- Management's value: 20% of the residual exit value:

  (0.20)($1,000 − $150 − $300) = $110.

Total initial investment by the private equity firm is $145, and by management $5.

Total payoff to the private equity (PE) firm at exit is $440 + $300 = $740.

Payoff multiple for the PE firm is $740 / $145 = 5.10.

Total payoff to management at exit is $110.

Payoff multiple to management is $110 / $5 = 22.0.

(Module 33.1, LOS 33.d)

Question #4 of 59

Question ID: 1473828

In a private conversation with his best friend, Harry Veeslay, CFA, makes the following statements:

Statement 1:	Private equity (PE) firms generally focus on short-term results. For example, they frequently use restructuring of acquired companies in an effort to quickly divest them for a profit.
Statement 2:	PE firms also want to ensure that the interests of portfolio company managers and of limited partners are aligned. For example, they frequently tie manager compensation to firm performance and include *tag-along*, *drag-along* clauses to give management a stake in the firm under certain trigger events.

With regard to Veeslay's statements:

A) both are incorrect. ✗

B) both are correct. ✗

C) only one is correct. ✓

Explanation

Statement 1 is incorrect. PE firms tend to have a long-term, rather than short-term focus in their investment strategies, which often exceeds 10 years. Restructuring is generally a lengthy process and requires a long-term perspective.

Statement 2 is correct with regard to both manager compensation and the use of *tag-along, drag-along clauses.*

(Module 33.1, LOS 33.b)

Question #5 of 59

Question ID: 1473877

An analyst is considering the performance of two private equity funds, Delta and Kappa.

Performance of private equity fund Delta and Kappa		
	Delta	**Kappa**
DPI	2.0	0.0
RVPI	0.0	2.0
TVPI	2.0	2.0

The *most appropriate* conclusion an analyst can draw from the table is that:

A) Kappa may be a younger fund than Delta. ✓

B) Kappa has distributed $2.0 for every dollar invested. ✗

C) Delta has yet to turn a profit. ✗

Explanation

Delta's distributed to paid-in capital (DPI) ratio of 2.0 indicates that investors in the fund realized a profit of $2.0 for every dollar invested and that this profit has already been paid out. Kappa's multiples indicate that the fund has yet to pay out profits to its investors. The residual value to paid-in capital (RVPI) of 2.0 implies that all returns are still unrealized and will be paid out in future years. One likely explanation for Kappa's multiples is that the fund is younger than Delta.

(Module 33.2, LOS 33.h)

Question #6 of 59

Question ID: 1473825

Norah Cyly is the recently appointed manager of a private equity fund that invests exclusively in venture capital investments in online fashion and media advertising companies. In a discussion with the fund's assistant portfolio manager, Cyly makes the following statements on control mechanisms and exit routes:

Statement 1:	*Earn-outs* are mainly used in venture capital investments. They relate the acquisition price paid by the limited partners to the future performance of the portfolio companies.
Statement 2:	It is generally difficult to value venture capital investments using the portfolio companies' cash flows or EBIT or EBITDA growth, since both cash flows and earnings are difficult to predict with certainty.

With respect to her statements, Cyly is:

A) correct on Statement 2 only.

B) correct on Statement 1 only.

C) correct on both statements.

Explanation

Both of Cyly's statements are correct. Her description of earn-outs as a control mechanism is accurate. Her comment on cash flows and earnings growth is also correct, given most venture capital firms' lack of stable cash flow and earnings patterns. This type of valuation is better suited for leveraged buyout investments.

(Module 33.1, LOS 33.b)

Question #7 of 59

Question ID: 1473855

The Dragonhill Group manages a $250 million private equity fund. Investors committed to a total of $300 million over the term of the fund and specified carried interest of 20% and a hurdle rate of 10%. Carried interest is distributed on a deal-by-deal basis. 60% of the $250 million has been invested at the beginning of year 1 in Deutsch Co. (Deutsch), with the remaining 40% invested in Reiner Ltd (Reiner).

Both firms are sold at the end of the third year, realizing a $45 million profit for Deutsch and a $35 million profit for Reiner.

The carried interest paid to the fund's general partner after Deutsch and Reiner, respectively, is:

	Deutsch	Reiner
A)	$9 million	$0

B)	$9 million	$7 million
C)	$0	$7 million

Explanation

Since carried interest is paid on a deal-by-deal basis, profits are not netted. Also, carried interest is only paid if the investment's IRR at least meets the hurdle rate of 10%.

(All figures are in $ million):

The initial allocation between the firms was:

Deutsch: (0.60)($250) = $150

Reiner: (0.40)($250) = $100

The IRRs for the two firms are:

IRRDeutsch: PV = -$150; FV = $195, N = 3; CPT I/Y → IRR = 9.14%.

IRRReiner: PV = -$100; FV = $135; N = 3; CPT I/Y → IRR = 10.52%.

Since the return on Deutsch fell short of the 10% hurdle rate, the general partner only receives profits after Reiner. The profit is 20% of $35 million, or $7 million.

(Module 33.2, LOS 33.g)

Question #8 of 59

Question ID: 1473841

The private equity firm Purcell & Hyams (P&H) is considering a $17 million investment in Eizak Biotech. Eizak's owners firmly believe that with P&H's investment they could develop their "wonder" drug and sell the firm in six years for $120 million. Given the project's risk, P&H believes a ROI of 4.83x is reasonable.

The pre-money valuation (PRE) and P&H's fractional ownership, respectively, are *closest* to (in millions):

	PRE	Fractional ownership
A)	$7.85	0.14
B)	$7.85	0.68
C)	$24.86	0.68

Explanation

Step 1: Calculate the post-money (POST) valuation (all dollar figures in millions):

POST = ($120) / 4.83 = $24.85 million.

Step 2: The pre-money valuation is Eizak's current value without P&H's investment:

PRE = $24.85 million − $17 million = $7.85 million.

Step 3: P&H's fractional ownership is the value of its investment as a fraction of Eizak's POST valuation:

f = INV / POST = $17 / $24.85 = 0.68.

(Module 33.1, LOS 33.d)

Jon Lester is considering diversifying his personal portfolio away from traditional markets by taking on some alternative investments, and is seeking advice. Lester has $150 million in his portfolio and so as a qualifying investor in the U.S. he is considering an opportunity to invest in a private equity fund, Titan Investments.

Titan has a range of investments in both young and old companies, and has often used high leverage to purchase young companies from the existing share holders. Lester has been invited by some of Titan's senior management to discuss a potential investment and a result has seen the reported performance on some of Titan's recent transactions.

Of particular interest to Lester was the method used to report the performance of the fund. To his knowledge it is recommended that PE firms use since inception IRR, and he was encouraged to see that Titan has adhered to this recommendation, particularly as it suits the liquidity position that private equity firms typically find themselves in.

In addition to the total return however, Lester is interested in the split of realized and unrealized gains investors have experienced over the life of the fund. He has been presented with a range of multiples prepared by the fund but is unsure how to interpret them. He has listed out the following notes and is seeking advice as to the best interpretation.

Financial Performance Metrics – Titan

Metric One: Paid in capital %

Metric Two: Distributed to PIC

Metric Three: Residual value to PIC

Lester also noted down some key figures regarding a venture capital investment which the fund is currently considering. The investment required by the portfolio company would be $12 million, and they would be looking for an ROI of 6x in 5 years' time when the portfolio company is expected to go public at a total valuation of $120 million. Initial discussions have apparently taken place, and the founders along with the current management team wish to keep their two million shares.

Question #9 - 12 of 59

Question ID: 1473872

Is Lester correct in his interpretation of performance reporting requirements for private equity firms?

A) Yes.

B) No. Although the method suits the liquidity position as it assumes only realized gains are reinvested, it is not recommended due to its complexity.

C) No. Although it is recommended it does not suit the liquidity position of PE firms as it assumes funds are reinvested at the IRR and the fund is often illiquid.

Explanation

GIPS recommend since inception IRR (essentially a yield measure) but it is assumption that PE firms can reinvest at the IRR is problematic. It assumes the fund is full liquid, whereas significant portion of the NAV is illiquid during a substantial part of the private equity funds life.

(Module 33.2, LOS 33.h)

Question #10 - 12 of 59

Question ID: 1473873

Which of Lester's multiples should he use in order to calculate the realized, unrealized, and total return on the fund?

A) Metric One gives total return, Multiple Two gives realized returns and Metric Three gives unrealized returns.

B) Metric Two gives realized returns and Metric Three gives unrealized returns. They should be added together to get total return.

C) Metric One gives total return, Multiple Two gives unrealized returns and Metric Three gives realized returns.

Explanation

DPI gives distributed, or realized returns, and RVPI gives residual value, or unrealized return. They should be added together to give total return.

(Module 33.2, LOS 33.h)

Question #11 - 12 of 59

Question ID: 1473874

Which of the following is *closest* to the pre-money valuation for the Venture Capital investment, and the fractional ownership required by Titan?

A) $8 million and 60%.

B) $20 million and 60%.

C) $12 million and 40%.

Explanation

Given: Exit value = $120m, ROI = 6x.

Post-money = 120/6 = $20m

Fractional ownership = 12/20 = 60%

Pre-money = POST – INV = 20 – 12 = $8m

(Module 33.1, LOS 33.d)

Question #12 - 12 of 59

Question ID: 1473875

Which of the following is *most* accurate regarding exit routes for PE investments?

A) An IPO is likely to lead to the highest price, is flexible but also costly.

B) An IPO is likely to lead to the highest price but is costly and not flexible.

C) An IPO is likely to lead to the lowest price but has a low cost.

Explanation

IPOs (Initial Public Offerings) have the following advantages:

- Higher valuation multiples via enhanced liquidity
- Access to capital
- Ability to attract higher caliber managers

IPOs have the following disadvantages:

- Expensive process (advisors, underwriters, etc.)
- Less flexible (timing of the IPO depends on the state of the financial markets)

(Module 33.2, LOS 33.e)

Question #13 of 59

Question ID: 1473843

The Milat Private Equity Fund (Milat) makes a $15 million investment in a promising venture capital firm. Milat expects the venture capital firm could be sold in four years for $150 million and determines that the appropriate ROI is 4x. The founders of the venture capital firm currently hold 1 million shares. Milat's fractional ownership in the firm and the appropriate share price, respectively, is *closest* to:

	Fractional ownership	Share price
A)	89.64%	$3.63
B)	40%	$22.50
C)	60%	$15.00

Explanation

The calculation requires four steps:

Step 1: Calculate POST:

POST = exit value / ROI = 150 million / 4 = $37.50 million

Step 2: Milat's fractional ownership of the venture capital firm is the investment divided by POST:

f = $15 million / $37.50 million = 0.40, or 40%

Step 3: Calculate the number of shares required by Milat (S_{pe}) for its fractional ownership of 40%:

S_{pe} = 1,000,000 [0.40 / (1 – 0.40)] = 666,667

Step 4: The share price is the value of Milat's initial investment divided by the number of shares Milat requires:

P = INV_1 / S_{pe} = $15 million / 666,667 = $22.50

Alternatively, PRE = POST – INV = 37.50 – 15 = 22.50 million

Price per share (founders) = 22.50 million / 1 million = $22.50

(Module 33.1, LOS 33.d)

Steve Squire, CFA, works for Opportunity Investments, a U.S.-based investment firm, which advertises a high level of expertise in the area of alternative investments. Steve is currently analyzing a request from a private equity firm to help with the analysis of a company, which it may invest in.

The target portfolio company, Meta Probes, is a high tech engineering company, which specializes in the production of scientific control apparatus designed to monitor the performance of man and machines in extreme environments.

It has recently fallen into financial difficulty after a major client went bankrupt leaving Meta Probes with a large amount of bespoke equipment, which had to be written off. Since the write off it has struggled to recover as the downturn in the economy has led to a drop off in research companies undertaking major projects.

The private equity firm is hoping to raise debt to finance a leveraged buy-out, which will also involve some of the existing management and skilled staff. The firm believes that Meta Probes still has a solid skill base and the ability to generate steady cash flows if it can restructure its capital base and see out the last half of the year.

It is hoped that an exit can be made in six years at a multiple of 1.95 of the firm's initial cost of $420 million. The firm is seeking assistance from Squire in analyzing this valuation and interpreting the components of the portfolio company's performance over the six years until the possible exit.

Squire has been provided with the following details on the potential deal. The initial investment is to be financed with 60% debt and 40% equity.

One suggested structure for the equity investment is as follows:

- $120 million in preference shares held by the private equity firm. The preference shares are promised an 8% compound annual return paid on exit
- $45 million in equity held by the private equity firm
- $3 million in equity held by management

The equity of the private equity firm is promised 90% of the residual value at exit after creditors and preference shares are paid. Management equity will receive the other 10% residual value.

By exit, it is estimated that the firm will have paid off $100m of the initial debt using operating cash flow.

In addition to the $3 million equity investment by top-level management, the private equity firm also wishes to include control mechanisms on the term sheet, and is seeking guidance from Squire in this area.

Question #14 - 17 of 59

Question ID: 1473858

Which of the following is not a typical source of value creation in private equity investments?

A) The ability to re-engineer the firm and operate it more efficiently. ✗

B) The ability to obtain debt financing on more advantageous terms. ✗

C) The separation of interests between management and private equity ownership to allow management to focus on core skills. ✓

Explanation

A source of value creation is the alignment of interests, not the separation.

(Module 33.1, LOS 33.d)

Question #15 - 17 of 59

Question ID: 1473859

Which of the following is Squire *least* likely to suggest as a possible control mechanism to include on the term sheet?

A) A tag-along, drag-along clause.

B) Performance related compensation.

C) Autonomy for management to make strategic changes to the business.

Explanation

The PE firm is likely to implement a Required Approvals clause, which would mean the PE firm must approve any strategic changes to the business.

(Module 33.2, LOS 33.g)

Question #16 - 17 of 59

Question ID: 1473860

The PE firm has asked Squire to assist them in using the Leveraged Buy Out (LBO) model to determine the return the firm should expect from the transaction. Which of the following is *closest* to the IRR that the PE Firm can expect from this deal?

A) 11.5%.

B) 24.7%.

C) 58.6%.

Explanation

The exit value for the deal is expected to be 1.95 × \$420m = \$819m in 6 years' time

The deal (\$420m) is 60% debt = \$252m and 40% equity = \$168m

At exit, the creditors will have debt outstanding of \$252m – \$100m = \$152m

The preference shares are promised an 8% return so their claim on value will be \$120m × $(1.08)^6$ = \$190.42m

Hence, the residual value available at exit for equity holders is \$819m – \$152m – \$190.42m = \$476.58m

The private equity firm is promised 90% (\$428.92m) of this \$476.58m and the management receive the remaining 10% (\$47.66m)

The total investment by the private equity firm at the start is \$120m + \$45m = \$165m

The total payoff to the private equity firm at exit is \$190.42m + \$428.92m = \$619.34m

Hence the IRR expected for the PE Firm is $[(619.34 / 165)^{(1/6)}] – 1$ = 24.7%

(Module 33.1, LOS 33.d)

Question #17 - 17 of 59

Question ID: 1484389

Which of the following is *closest* to the IRR that the senior management can expect from this deal?

A) 11.5%.

B) 26.2%.

C) 58.6%.

Explanation

Management initially invest \$3m equity.

Using the figures from question 3, the exit value for management is expected to be \$47.66m.

The IRR is therefore $(47.66 / 3)^{(1/6)} – 1$ = 58.6%

(Module 33.1, LOS 33.d)

Question #18 of 59

Question ID: 1473822

The Jefferson Group is a large private equity firm managing a multi-billion dollar portfolio. Which of the following is the *least likely* source of value-added the Jefferson Group would provide to its portfolio companies (as compared to a public firm)?

A) Obtaining cheap credit.

B) Aligning the interests between private equity owners and limited partners.

C) Reengineering the portfolio companies.

Explanation

The three sources of value-added a private equity firm provides over public firms are: reengineering the portfolio firms, obtaining debt on favorable terms (cheap credit), and aligning the interests between private equity owners (the limited partners) and portfolio managers.

(Module 33.1, LOS 33.a)

Question #19 of 59

Question ID: 1473837

Which of the following statements is the *least appropriate*?

A) Debt amortization in a leveraged buyout investment increases risk to the investor as it is a burden on the firm's cash flow.

B) Leverage in a leveraged buyout investment can be advantageous as debt amortization can magnify investor returns.

C) Leverage in a leveraged buyout investment can be disadvantageous as debt increases risk to the investor if the firm cannot meet its interest obligation.

Explanation

As the amortization of debt reduces investor risk (less debt outstanding) and the reduced claim by debtholders can actually magnify investor returns.

(Module 33.1, LOS 33.d)

Question #20 of 59

Question ID: 1473881

Which of the following pairs *correctly* identifies the fees paid to agents for raising funds for the private equity firm, and the fees paid to the general partner (GP) for investment banking services, respectively?

Fees to agents Fees to GP

A)	Transaction fees	Administrative costs	✗
B)	Administrative costs	Placement fees	✗
C)	Placement fees	Transaction fees	✓

Explanation

Placement fees are upfront fees paid to agents for raising funds for the private equity firm. These fees typically are in the 2% range or paid as trailers.

Transaction fees are paid to the GP for investment banking services in the event of a merger or acquisition. Transaction fees are usually split with the limited partners and deducted from management fees.

Administrative costs are various annual costs including custodian fees, fees to transfer agents and accounting costs.

(Module 33.3, LOS 33.i)

Question #21 of 59

Question ID: 1473854

RDO is a private equity fund with $50 million in committed capital and an investment in three portfolio companies totalling $30 million. The fund earned a healthy profit of $5 million after its first year on the sale of one of the companies but suffered a $2 million loss after its second year on the sale of the second company. The fund pays carried interest of 20% on a *total return basis* using committed capital and also has a clawback provision.

The clawback the general partner must pay at the end of the second year is:

A) $600,000. ✗

B) $0. ✓

C) $400,000. ✗

Explanation

A clawback provision in a private equity prospectus requires the general partner to repay part of previously distributed profits if the fund subsequently underperforms.

Since carried interest is paid on a total return basis using committed capital, the general partner of RDO would only receive interest when the portfolio value exceeds committed capital ($50 million). First-year profit is $5 million, bringing the portfolio value to $35 million, therefore no carried interest is paid. Since no profit was distributed to the general partner in the first year, a clawback does not apply in the second year.

(Module 33.2, LOS 33.g)

Question #22 of 59

Question ID: 1473876

The relevant measure of cash flows for the limited partners (LPs), and the LPs' realized return from investment in the private equity fund, respectively, is:

	Return metric	LPs' realized return
A)	Paid-in capital	Net IRR
B)	Gross IRR	Residual value to paid-in capital
C)	Net IRR	Distributed to paid-in capital

Explanation

Net IRR measures the cash flows between the fund and the limited partners and is therefore the relevant return metric for the LPs. Distributed to paid-in capital (DPI) measures the LPs' realized return from investment in the fund. It is calculated as the cumulative distributions already paid to the LPs over the cumulative invested capital.

Gross IRR measures the cash flows between the fund and the portfolio companies. Residual value to paid-in capital (RVPI) measures the LPs' unrealized return from the fund. Paid-in capital measures the percent of capital used by the general partner.

(Module 33.2, LOS 33.h)

Question #23 of 59

Question ID: 1473830

The *most appropriate* pairing for valuing a buyout and a venture capital investment, respectively, is:

	Buyout	Venture capital
A)	Relative value approach	Discounted cash flow
B)	Discounted cash flow	Pre-money valuation

C)	Pre-money valuation	Relative value approach

Explanation

Buyout investments have predictable cash flows and there are typically several comparable firms in the industry. Both the discounted cash flow and relative value approach are thus reasonable valuation techniques for buyout firms.

Venture capital firms, on the other hand, have less stable cash flows and few industry comparables given their young age and position in the business life cycle. Pre- and post-money valuation techniques are frequently used valuations for these firms.

(Module 33.1, LOS 33.c)

Question #24 of 59

Question ID: 1473845

Mavis Krager, manager of alternative investments for the Richmond Group, is considering the merits of some private-equity opportunities. Richmond Group likes to invest in private-equity funds, but will also do its own deals if the opportunity is right. One deal on the table is an equity stake in Melton Motors, a chain of privately held auto dealerships. The company is well run, but has come upon hard times lately because of credit problems. Krager thinks Melton will solve its financial problems and become profitable again. She is considering investing $7 million in the company. Also under discussion is The Apple House, a large privately held orchard in Wisconsin. Richmond Group is considering investing $1 million.

To determine whether the deals are worthwhile Krager decides to estimate a price for each company based on a post-money valuation. Apple House offer is $25.00 per share and Melton Motors offer price is $2.50 per share. The investment firm prefers to focus on companies willing to price their stocks at least 20% below their true value and fund the investments only once. To calculate her valuations, Richmond uses the data below:

	Melton Motors	The Apple House
Required ROI	5X	10X
Number of shares held by current owners	1.5 million	80,000
Estimated value of company at end of investment period	$51 million	$29 million
Expected length of investment	5 years	10 years

Just as Krager finishes her assessment of the two private-equity deals, a contact at The Apple House calls her and says the management team is considering a leveraged buyout (LBO) and wants Richmond Group to help finance it. Since the firm hasn't financed an LBO for years, Krager gets out a book she has not read since college to bone up on the valuation equations and reacquaint herself with terms specific to LBOs.

What action should Richmond Group take with regard to:

	Melton Motors	The Apple House
A)	Buy stake	don't buy stake
B)	Don't buy stake	don't buy stake
C)	Don't buy stake	buy stake

Explanation

Step 1: Calculate the post-money valuation (POST).

POST = exit value / ROI

POST for Melton = $51 million / 5 = 10.2 million.

POST for Apple = $29 million / 10 = 2.9 million.

Step 2: Calculate pre-money valuation.

PRE = POST − investment.

PRE for Melton = 10.2 million − $7 million = 3.2 million.

PRE for Apple = 2.9 million − $1 million = 1.9 million.

Step 3: Determine the fractional ownership.

F = INV / POST

F for Melton = $7 million / 10.2 million = 68.63%.

F for Apple = $1 million / $2.90 million = 34.48%.

Step 4: Determine the number of shares the firm must buy.

Stake = Entrepreneurs' shares × [F / (1 − F)].

Stake for Melton = 1.5 million * (0.6863 / 0.3137) = 3,281,639 shares.

Stake for Apple = 80,000 * (0.3448 / 0.6552) = 42,100 shares.

Step 5: Calculate stock price per share.

P = INV / Stake

P for Melton = $7 million / 3,281,639 = $2.13

P for Apple = $1 million / 42,100 = $23.75

Both valuations result in a price below the offer price. Richmond Group should not buy either stake.

(Module 33.1, LOS 33.d)

Question #25 of 59

Question ID: 1473880

The Nishan private equity fund was established five years ago and currently has a paid-in capital of $300 million and total committed capital of $500 million. The fund paid its first distribution three years ago of $50 million, $100 million the year after and $200 million last year. The fund's distributed to paid-in capital (DPI) multiple is *closest* to:

A) 0.67.

B) 1.17.

C) 0.70.

Explanation

The DPI multiple is calculated as the cumulative distributions paid by the private equity fund divided by the paid-in capital (the portion of committed capital drawn down).

Nishan's current DPI is: ($50 + $100 + $200) / $300 = 1.17

(Module 33.3, LOS 33.i)

Sammy Porter is a qualified investor based in the U.S. He is currently looking to invest up to $5 million and has decided to consider private equity funds as a potential vehicle. Porter has looked into several options and met with Michael Weber, an ex-colleague of Porter's who has several years' experience in the world of private equity.

Weber has extended an invitation to Porter to visit his offices in and discuss a prospectus for a potential new venture, which Weber is in the process of setting up.

Porter has reviewed the prospectus and highlighted several areas, which he wished to discuss with Weber. The points he has highlighted include the following:

Point 1

Under the terms of the prospectus, Porter would invest as a Limited Partner in a Limited Partnership, with Weber as the General Partner. Porter is unclear as to the personal liability that results from such an arrangement.

Point 2

Carried interest is to be calculated as 20% after management fees using the deal-by-deal method. The appendix to the prospectus includes examples of how this would be calculated. One example details a theoretical fund, which has a carried interest rate of 20%. An investment of $30 million is made and later in the year, the fund exits the investment and earns a profit of $18 million.

Point 3

Another section of the prospectus mentions some of Weber's previous investments and the use of ratchet mechanisms. The prospectus mentions that typically ratchet mechanisms affect the allocation of equity between shareholders and managers. A ratchet allows the shareholders to increase their equity allocation depending on the company's performance.

Porter also intends to question Weber on venture capital investments. Two entrepreneurs who run a new start software company have approached Porter. The startup had originally raised financing of $2 million with an expectation of going public at a valuation of $45 million. Round 1 investors had priced in ROI of 9x. The firm now needs an additional capital infusion of $10 million and the firm's IPO valuation has been revised to $200 million. Porter is aware of the risk involved and would use a ROI of 5x.

Question #26 - 29 of 59

Question ID: 1473863

Regarding Point 1 that Porter makes, which of the following is *most* accurate?

A) Porter's liability would be limited to his investment, Weber's would be unlimited. ✓

B) Weber's liability would be limited to his investment, Porter's would be unlimited. ✗

C) Porter and Weber both have limited liability as they are both partners in a limited liability partnership. ✗

Explanation

The limited partners have limited liability whereas the general partner has unlimited liability.

(Module 33.2, LOS 33.g)

Question #27 - 29 of 59

Question ID: 1473864

For the example given in Point 2, how much carried interest would Porter be entitled to after the fund exits the investment?

A) $3.6 million. ✗

B) $3.6 million if the total proceeds of $48 million are more than the committed capital of the fund. ✗

C) None. ✓

Explanation

The carried interest is the General Partner's share of fund profits and Porter is to be a limited partner.

(Module 33.2, LOS 33.g)

Question #28 - 29 of 59

Question ID: 1473865

Regarding Point 3, are the comments relating to ratchet mechanisms *correct*?

A) Correct. ✗

B) Incorrect, ratchet mechanisms increase the price the PE firm must pay for the portfolio company based on future performance. ✗

C) Incorrect, with regard to the shareholders increasing their allocation. ✓

Explanation

Ratchet mechanisms typically increase the equity allocation to management team of the portfolio company, based on performance metrics, rather than the shareholders of the PE firm.

(Module 33.2, LOS 33.g)

Question #29 - 29 of 59

Question ID: 1473866

Assuming that the round one investors do not participate in the second round of financing for the startup, the round one investors' fractional ownership after the second round is *closet* to:

A) 25%. ✗

B) 30%. ✓

C) 40%. ✗

Explanation

Given: INV_1 = \$2m, Exit $value_1$ = \$45m, ROI_1 = 9x, INV_2 = \$10m, Exit $value_2$ = \$200m, ROI_2 = 5x.

$POST_1$ = exit $value_1$ / ROI_1 = 45 / 9 = \$5m

f_1 = INV_1 / $POST_1$ = 2 / 5 = 0.40 or 40%

$POST_2$ = exit $value_2$ / ROI_2 = 200 / 5 = \$40m

f_2 = INV_2 / $POST_2$ = 10 / 40 = 0.25 or 25%

After the second round, the fractional ownership of the first round investors will decline by 25% to 0.40(1 – 0.25) = 0.30 or 30%

(Module 33.1, LOS 33.d)

Question #30 of 59

Question ID: 1473824

Contrary to most public companies, the magnitude that debt is typically utilized in private equity (PE) LBO transactions and the way this debt is quoted, respectively, is:

	Debt is utilized	Debt is quoted	
A)	more heavily	as a multiple of equity	✗

B) less heavily	as a multiple of sales	✗
C) more heavily	as a multiple of EBITDA	✓

Explanation

PE LBO transactions typically use higher leverage than most public companies do. Debt is usually quoted as a multiple of EBITDA, while public firm debt is usually quoted as a multiple of equity (debt-to-equity ratio).

(Module 33.1, LOS 33.a)

Question #31 of 59

Question ID: 1473867

An implicit cost in private equity of additional financing or issuing stock options to management is called:

A) management and performance cost. ✗

B) dilution cost. ✓

C) capital cost. ✗

Explanation

Management and performance cost is the explicit cost of manager compensation as a percentage of committed capital and annual fund performance. Capital costs are not discussed as a cost in private equity.

Dilution is the implicit cost of reduced investor value when firms take on additional financing or when stock options are granted (and exercised) by management.

(Module 33.2, LOS 33.f)

Question #32 of 59

Question ID: 1473850

Christina Wagner is a CFA level II candidate currently studying about hedge funds, private equity and commodity futures. One of her friends is fascinated by what Wagner is learning and asks several questions on the topic. In particular, she is curious to know what exit options are available to a promising young venture capital (VC) firm if it is having difficulty attracting buyers due to poor market conditions. What should be Wagner's *most appropriate* response?

A) Since an initial public offering is not feasible, the VC firm should be sold to another firm through a buyout or secondary market sale.

B) The VC firm should be liquidated in the absence of prospective buyers through the sale of the firm's assets.

C) The VC firm should consider the acquisition of another firm and sell the merged entity once capital market conditions have improved.

Explanation

Liquidation occurs when a firm becomes insolvent or bankrupt, cannot function as an independent entity, and there are very few or no interested buyers. Liquidation results in low exit values. Selling the VC firm through a buyout or secondary market sale is also less feasible since these transactions require significant debt financing which the young VC firm may be unable to support.

In poor market conditions it may be feasible for the VC firm to make a strategic acquisition through a merger and sell the merged entity once market conditions have stabilized.

(Module 33.2, LOS 33.e)

Question #33 of 59

Question ID: 1473869

An analyst makes the following statements on the risk and costs of private equity investments:

Statement 1:	Committed capital is the initial capital in a private equity fund to obtain first round financing. As committed capital is used up, investors are required to make additional commitments to finance firm projects and expansion.
Statement 2:	The J-Curve refers to the risk pattern in a private equity investment over time. Risk in private equity investments initially typically declines as more capital is drawn down but increases closer to exit since exit timing and values are difficult to predict.

With respect to the analyst's statements:

A) both are correct.

B) both are incorrect.

C) only one is correct.

Explanation

Both statements are incorrect. Committed capital refers to the amount of funds investors committed to over the life of the private equity fund. Funds from committed capital are drawn down over time as the firm needs more capital. If the firm needs financing beyond investors' committed capital, it would have to look for additional sources of funds.

The J-Curve refers to a pattern in private equity investment return, not risk. The return on investments usually declines initially, then increases as exit nears.

(Module 33.2, LOS 33.f)

Question #34 of 59

Question ID: 1473831

A private equity investor is considering making an investment in a venture capital firm. The investor values the firm at $1.5 million following a $300,000 capital investment by the investor. The venture capital firm's pre-money (PRE) valuation and the investor's proportional ownership, respectively, are:

	PRE valuation	Ownership proportion
A)	$1.5 million	20%
B)	$1.2 million	20%
C)	$1.5 million	25%

Explanation

The post-money valuation (PRE) is directly given as $1.5 million. The pre-money valuation (PRE) is simply the post-money valuation (POST) less the investment (INV):

PRE = POST − INV = $1.5 million − $300,000 = $1.2 million.

The ownership proportion is the investor's fractional ownership of the firm value after the capital infusion:

Ownership proportion = INV/POST = $300,000 / $1.5 million = 0.20 or 20%.

(Module 33.1, LOS 33.d)

Question #35 of 59

Question ID: 1473832

A private equity firm is considering the valuation characteristics of both a venture capital and a buyout investment. Increasing working capital requirements and stable EBITDA growth is *most likely* associated with:

	Increasing working capital	Stable EBITDA growth
A)	Buyout	Venture capital
B)	Buyout	Buyout
C)	Venture capital	Buyout

Explanation

Venture capital investments often have high and increasing working capital (current assets less current liabilities) requirements to finance growth. Buyouts typically have low requirements due to more reliable cash flows and earnings and a substantial asset base.

Stable EBITDA (or EBIT) growth is generally a characteristic of buyout investments. These firms traditionally have a history of stable sales and cash flows and have already established a strong market position. The high amount of debt required by the private equity firm to make the investment also requires that the buyout firm have stable and steady earnings to finance the interest payments.

(Module 33.1, LOS 33.c)

Yanish Cheung, CFA, works for a U.S.-based institutional investor in private equity. He is reviewing the results of Bud Rosewood I, a venture capital fund specializing in U.S. high- technology companies in their start-up stage of development.

As the general partner (GP) of the fund comes with significant reputation, the fund is able to charge a management fee of 2.25% and a carried interest of 25% with a committed capital of $105m. The carried interest is computed using the first alternative of the total return method (i.e., the GP will first charge carried interest when NAV before Distributions exceeds committed capital). In subsequent years, provided NAV before Distributions exceeds committed capital, the GP will charge carried interest on the increase in NAV before Distributions.

Exhibit 1 summarizes the results of Bud Rosewood I for the past 6 years. Unfortunately, some of the numbers for 20X5 are missing.

Exhibit 1: Bud Rosewood I – Cash Flows and Distributions (USD million)

Year	Called-down Capital	Paid-in capital	Management Fee	Operating Results	NAV before Distributions	Carried Interest	Distributions	NAV after Distributions

20X0	45.0	45.0	(1.0)	(3.0)	41.0	0.0	0.0	41.0
20X1	12.0	57.0	(1.3)	(12.0)	39.7	0.0	0.0	39.7
20X2	8.0	65.0	(1.5)	13.0	59.2	0.0	0.0	59.2
20X3	24.0	89.0	(2.0)	30.0	111.2	(1.6)	(18.0)	91.7
20X4	7.0	96.0	(2.2)	40.0	136.5	(6.3)	(37.0)	93.2
20X5	4.0			60.0			(68.0)	

Cheung is very careful about evaluating the corporate governance features of the funds in which he invests. To this end, he asks a junior analyst, Roy Zograff, to look at two other funds. Information about them is summarized in Exhibit 2:

Exhibit 2

	Fund Alpha	**Fund Bravo**
Management Fee	3%	1%
Carried Interest	15%	30%
Hurdle Rate	5%	10%
Clawback Provision	No	Yes
Distribution Waterfall	Deal-by-Deal	Total Return
PIC	0.40	0.80
DPI	0.20	0.65
RVPI	1.25	0.95

Having collected the data, Zograff believes that the "cash-on-cash" return of Fund Alpha is lower, but its unrealized return is higher compared to that of Fund Bravo.

Based on the information contained in Exhibit 2, Cheung makes the following two comments:

Comment 1:	The interests of the GP and LPs are more likely better aligned in Fund Alpha.
Comment 2:	The GP in Fund Alpha will receive carried interest earlier than the GP of Fund Bravo, keeping all else constant.

Question #36 - 39 of 59

Question ID: 1473885

The DPI of Bud Rosewood I for 20X5 is *closest* to:

A) 1.23.

B) 1.33.

C) 1.43.

Explanation

Paid-in capital in 20X5 = paid-in capital 20X4 + capital called down in 20X5 = 96 + 4 = $100m

Total distributions are 18 + 37 + 68 = $123m.

So Distributions to paid-in capital = 123/100 = 1.23

(Module 33.3, LOS 33.i)

Question #37 - 39 of 59

Question ID: 1473886

The RVPI of Bud Rosewood I for 20X5 is *closest* to:

A) 0.79.

B) 0.82.

C) 0.85.

Explanation

We need the NAV after Distributions in 20X5. We first compute the management fee = 2.25% × $100m = $2.25m.

We compute NAV before Distributions = $93.2 + $4 – $2.25 + $60 = $154.95

Carried interest is 25% × increase in NAV before Distributions = 25% × (154.95 – 136.5) = $4.61m

NAV after Distributions = $154.95 – $4.6 – $68 = $82.35m

The residual value to paid-in capital (RVPI) = NAV after Distributions/paid-in capital = 82.35/100 = 0.82

(Module 33.3, LOS 33.i)

Question #38 - 39 of 59

Question ID: 1473887

How do you evaluate the comments made by Cheung?

A) Both comments are correct.

B) Only Comment 1 is correct.

C) Only Comment 2 is correct.

Explanation

Comment 1 is incorrect. Despite higher carried interest (30% vs. 15%), Fund Bravo has a higher hurdle rate and has a clawback provision. The hurdle rate indicates that carried interest will only be paid if the long-term returns of the fund exceed it. A clawback provision means that investors can claim back carried interest in the event of suboptimal performance of the fund over its life. The presence of both a higher hurdle rate and a clawback provision indicate that management of Fund Bravo have a stronger incentive to generate wealth for their investors.

Comment 2 is correct. Distribution waterfall structure based on the deal-by-deal method results in carried interest being paid to the GP after each successful exit. This method is mostly employed in the United States.

(Module 33.2, LOS 33.g)

Question #39 - 39 of 59

Question ID: 1473888

In relation to Fund Alpha, Zograff is right about:

A) both metrics.

B) only about the “cash-on-cash” return.

C) only about the unrealized return.

Explanation

Fund Alpha has a lower DPI but higher RVPI compared to Fund Bravo. Distributions to paid-in capital is a proxy of "cash-on-cash" return (or realized return) whereas residual value to paid-in capital is a proxy of unrealized return.

(Module 33.2, LOS 33.h)

Question #40 of 59

Question ID: 1473826

Private equity firms can maintain control over portfolio companies in a variety of ways. Which of the following contract terms would *least likely* achieve this goal?

A) Tag-along, drag-along clauses.

B) Priority in claims.

C) Board representation.

Explanation

A tag-along, drag-along clause is less a control mechanism for private equity firms and more a tool to tie portfolio manager interests to the portfolio companies. The clause gives portfolio managers the right to obtain an equity stake in the portfolio companies should the private equity firm decide to dispose of its holding.

Priority in claims and board representation are both effective tools that give PE firms greater control over portfolio companies. Priority in claims allows the PE firm to receive distributions before all other owners. Should the portfolio company experience a major event (bankruptcy, restructuring, IPO, etc.), the private equity firm can gain control of the company through board representation.

(Module 33.1, LOS 33.b)

Question #41 of 59

Question ID: 1473878

An investor in a private equity fund realizes that the residual value to paid-in capital (RVPI) is fairly large relative to the distributed to paid-in capital (DPI). The *most* appropriate conclusion drawn by the investor would be that:

A) it will take longer for the investor to realize a return from the fund.

B) there were significant cash flows from the fund to the investor.

C) the fund successfully earned profits from its investments.

Explanation

Paid-in capital measures the amount of capital drawn down out of total committed capital. Residual value to paid-in capital is the value of the investor's holding in the fund as a ratio of cumulative invested capital.

A high RVPI to DPI ratio indicates that the fund has not distributed a large portion of profits and may indicate difficulty realizing profits from its investments. In this case it would take longer for the investor to receive distributions from the fund (low cash flows to date).

(Module 33.3, LOS 33.i)

Question #42 of 59

Question ID: 1473882

A private equity fund pays a management fee of 3% of PIC and carried interest of 20% to the general partner using the total return method based on committed capital. In 2008 the fund has drawn down 80% of its committed capital of $250 million, and has a net asset value (NAV) before distributions of $260 million. The 2008 management fee and carried interest paid, respectively, is (in millions):

	Management fee	Carried interest
A)	7.8	2.0
B)	7.5	50.0
C)	6.0	2.0

Explanation

(All dollar figures are in millions)

Management fee is paid annually on paid-in capital (PIC), which is just cumulative capital drawn down. 2008 management fee is thus 3% of $200, or $6.0.

Carried interest is the profit distributed to the general partner. The fund specifies a total return method based on *committed* capital and is calculated as the excess of NAV before distributions above committed capital. The 2008 carried interest paid out is then 20% of ($260 − $250) = $2.0.

(Module 33.3, LOS 33.i)

Question #43 of 59

Question ID: 1473827

Pauler Investment Co. ("Pauler") just proposed to make a sizeable investment in Bada Cork, a recently established Hungarian producer of synthetic wine bottle corks with a patented new technology. Pauler is looking to make further strategic acquisitions in small venture capital companies in the food and beverage industry and has set up a fund to manage the portfolio companies. It has also brought onboard Kristina Sandorf as portfolio manager. Upon receiving her contract, Sandorf complains to a friend of the contract terms proposed by Pauler. In particular, she grumbles that an *earn-out* clause is inserted, which she believes would give Pauler priority on the earnings and dividends of companies in the portfolio ahead of herself.

In her description of *earn-outs*, Sandorf is:

A) incorrect, because earn-outs refer to Pauler having priority over Bada's assets in case of bankruptcy or liquidation.

B) incorrect, because earn-outs refer to tying the acquisition price paid by Pauler for the portfolio companies to the companies' future performance.

C) correct.

Explanation

Earn-outs are typically used in venture capital investments where the acquisition price paid for portfolio companies by private equity firms is tied to the companies' future performance.

(Module 33.1, LOS 33.b)

Question #44 of 59

Question ID: 1473840

The private equity firm Purcell & Hyams (P&H) is considering a $10 million investment in Eizak Biotech. Eizak's owners firmly believe that with P&H's investment they could develop their "wonder" drug and sell the firm in six years for $120 million. Given the project's risk, ROI of 10x is deemed appropriate. Four years after this investment, an additional round of investment of $7 million is needed (Series B shares). Series B investors agree that the exit will still occur in two years but at a much higher valuation of $150 million. ROI for Series B investors is 5x. The fractional ownership for P&H after the second round would be *closest* to:

A) 0.75.

B) 0.23.

C) 0.64.

Explanation

Step 1: Calculate $POST_1$:

$POST_1$ = exit $value_1$ / ROI_1 = 120/10 = 12

Step 2: Calculate fractional ownership f_1:

f_1 = INV_1 / $POST_1$ = 10/12 = 83.33%

Step 3: Calculate $POST_2$:

$POST_2$ = exit $value_2$ / ROI_2 = 150/5 = 30

Step 4: Calculate fractional ownership f_2:

f_2 = INV_2 / $POST_2$ = 7/30 = 23.33%.

Step 5: Fractional ownership of first round investors after the second round:

f_1 (post round 2) = f_1 (1 – f_2) = 0.8333 (1 – 0.2333) = 63.89%

(Module 33.1, LOS 33.d)

Question #45 of 59

Question ID: 1473838

Which of the following statements *most accurately* describes the components of returns on a leveraged buyout (LBO) investment:

A) The interest earned on debt financing, the return on common shares and the return on preference shares.

B) The return on common shares, the increase in the price multiple on exit, and the equity held by management.

C) The return on preference shares, the increase in the price multiple on exit, and the reduction in debt claims.

Explanation

The components of a private equity firm's returns are the return on preference shares, the increased price multiple and the reduction in debt claims. The private equity firm should see an increase in the price multiples as the operational efficiencies of the LBO firm improve. The second component is the value of the interest-bearing preference shares. The third component is the reduction in debt over the time period to exit.

(Module 33.1, LOS 33.d)

Question #46 of 59

Question ID: 1473856

The *most relevant* market risk to a private equity investor is:

A) both short-term and long-term macro changes.

B) long-term macro changes only.

C) short-term macro changes only.

Explanation

Private equity investments are affected to a large degree by long-term macro- factors such as interest rate and exchange rate fluctuations and various market risks. Short-term macro-factors and short-term fluctuations are less relevant as the investor's time horizon typically exceeds 10 years.

(Module 33.2, LOS 33.g)

Question #47 of 59

Question ID: 1473842

A private equity firm makes a $10 million investment in a portfolio company. The founders of a portfolio company currently hold 300,000 shares and the pre-money valuation is $6 million. The number of shares to be held by the private equity firm, and the appropriate share price, respectively, are *closest* to:

	Number of shares	Share price
A)	500,000	$32.00
B)	500,000	$20.00
C)	480,000	$20.83

Explanation

The answer requires four steps:

Step 1: Calculate the post-money (POST) valuation, which is simply the pre-money (PRE) valuation plus the investment:

POST = PRE + INV = $6 million + $10 million = $16 million

Step 2: Calculate the private equity firm's fractional ownership in the portfolio company:

f = INV / POST = $10 million / $16 million = 0.625

Step 3: If the founders currently hold 300,000 shares, the number of shares to be held by the private equity firm to have 62.5% ownership is:

Number of shares = 300,000 [0.625 / (1-0.625)] = 500,000

Step 4: Given the private equity firm's $10 million investment and 500,000 shares, the share price is calculated as:

P = $10 million / 500,000 = $20.00

(Module 33.1, LOS 33.d)

Question #48 of 59

Question ID: 1473851

Which of the following is the *least likely* disadvantage in calculating the net asset value (NAV) for a private equity fund?

A) The limited partners use a third party to calculate the NAV of a private equity fund. ✓

B) NAV may be difficult to calculate since firm values are not known with certainty prior to exit. ✗

C) Only capital commitments already drawn down are included in the NAV calculation. ✗

Explanation

NAV is usually calculated by the fund's general partner, which could result in a subjective and inflated NAV. Limited partners, however, often use third party valuations to arrive at an objective and up-to-date NAV. This scenario thus describes a countermeasure to an issue in calculating NAV rather than a disadvantage itself.

The other two answers are both disadvantages in calculating NAV.

(Module 33.2, LOS 33.g)

Question #49 of 59

Question ID: 1473849

Which of the following lists *correctly* identifies exit routes in private equity, arranged from lowest to the highest exit values?

A) Liquidation, secondary market sale, IPO. ✓

B) Management buyout, liquidation, IPO. ✗

C) Initial public offering (IPO), management buyout, secondary market sale. ✗

Explanation

Liquidation is a sale of last resort for bankrupt or insolvent firms and generally results in low exit values. The value realized on the sale to management in a management buyout typically varies, but lags behind values from a secondary market sale or an IPO.

A secondary market sale is analogous to a private sale of the firm to another firm. Secondary market sales use large amounts of debt financing and could result in the second highest valuation after an IPO. An IPO is a sale of the entire firm or part of the firm (e.g. a division) to the public. As a result of the increased post-IPO liquidity, transparency and access to capital, the private equity firm can realize the highest exit value of a firm through the IPO process.

(Module 33.2, LOS 33.e)

Question #50 of 59

Question ID: 1473848

The primary advantage of an initial public offering (IPO) as an exit route in private equity is that it:

A) offers the highest exit value potential. ✓

B) is more cost-efficient and flexible than alternative exit routes. ✗

C) is appropriate for firms regardless of firm size and operating history. ✗

Explanation

A private equity firm can generally realize the highest exit value for a portfolio company through an IPO, as the post-IPO firm offers greater liquidity (it is continuously traded on an open exchange) and access to capital. IPOs, however, are costly to implement and involve a complex process that ranges from dealing with underwriters, gauging market interest and complying with various regulatory requirements. IPOs are also most appropriate for large firms with a stable operating history.

(Module 33.2, LOS 33.e)

Question #51 of 59

Question ID: 1473879

The net asset value (NAV) *after* distributions of a private equity fund is calculated as:

A) NAV before distributions + Carried interest – Distributions.

B) NAV before distributions + Capital called down – Management fees.

C) NAV before distributions – Carried interest – Distributions.

Explanation

NAV after distributions is calculated as NAV before distributions minus carried interest (the general partner's profit from the fund) minus distributions from the fund.

(Module 33.3, LOS 33.i)

Question #52 of 59

Question ID: 1473829

Analysts Jordan Green and Noelle Lafonte are discussing terminal value estimation in venture capital and buyout investments.

Lafonte states: "Private equity firms often use scenario analysis in both venture capital and buyout investments to estimate terminal value."

Green adds: "Private equity firms only use the multiple of net income approach in leveraged buyout (LBO), but not in venture capital investments to estimate terminal value."

With respect to their statements:

A) Neither Lafonte nor Green is incorrect.

B) Green is correct but Lafonte is incorrect.

C) Lafonte is correct but Green is incorrect.

Explanation

Lafonte's statement is correct. Private equity firms can use scenario analysis to estimate terminal value in both venture capital and LBO investments. Under scenario analysis, terminal values are calculated under multiple scenarios using different assumptions.

Green's statement is incorrect. Private equity firms often use a relative value approach to estimate terminal value in both venture capital and LBO investments. Under the multiple of net income approach, terminal year net income is multiplied by the P/E ratio to project terminal equity value.

(Module 33.1, LOS 33.c)

Question #53 of 59

Question ID: 1473868

Private equity values have declined significantly over the last year. Which of the following risk factors is the *least likely* reason for the decline?

A) Tax risk.

B) Market risk.

C) Investment-specific risk.

Explanation

Market risk is the risk of long-term changes in interest rates, exchange rates and economic risk. Certainly all of these have been factors in the less than spectacular private equity returns recently. Investment-specific risk is probably the most important source of risk in recent times, as many private equity investments suffered significant losses as a result of the subprime mortgage and real estate meltdown. Tax risk is the risk of tax changes over time, which has not been a significant factor in private equity valuations recently.

(Module 33.2, LOS 33.f)

Question #54 of 59

Question ID: 1473846

A private equity investor expects to realize a return on her venture capital investment in two years and expects to sell the firm for $30 million. She expects to make an investment of $5 million at an ROI of 2x. The post-money value of her investment today is *closest* to:

A) $60 million.

B) $10 million.

C) $15 million.

Explanation

POST = exit value / ROI = 30 / 2 = 15 million

(Module 33.1, LOS 33.d)

Question #55 of 59

Question ID: 1473847

The founders of a small technology firm are seeking a $1 million venture capital investment from prospective investors. The firm is expected to have revenues of $15 million in four years and comparable firms are valued at 2X revenues. ROI of 10x is deemed appropriate given the risk of the investment.

The pre-money valuation (PRE) of the technology firm is *closest* to (in millions):

A) $3.45.

B) $2.33.

C) $2.00.

Explanation

Exit value = revenues × 2 = 15 × 2 = $30 million

POST = exit value / ROI = 30 / 10 = $3 million

PRE = POST – INV = $3 – $1 = $2 million

(Module 33.1, LOS 33.d)

Question #56 of 59

Question ID: 1473852

The pair of terms that *correctly* identifies the method of profit distribution between limited partners (LPs) and general partners (GPs), and the allocation of equity between shareholders and management of a portfolio company, respectively, is:

	Method of profit distribution	Equity allocation	
A)	Ratchet	Carried interest	✗
B)	Distribution waterfall	Ratchet	✓
C)	Carried interest	Distribution waterfall	✗

Explanation

Distribution waterfall identifies the profit allocation between LPs and GPs and specifies when GPs can receive carried interest. *Ratchet* refers to the equity allocation between shareholders and management. *Carried interest* is the GP's share in fund profits.

(Module 33.2, LOS 33.g)

Question #57 of 59

Question ID: 1473883

Dr. Jason Bruno is a qualified investor in the US who is considering a $10 million investment in a private equity fund. Upon reading the fund's prospectus, Dr. Bruno encounters several contract terms and expressions with which he is unfamiliar. In particular, he would like to know the meaning of *ratchet* and distributed paid-in capital (*DPI*). The *most appropriate* answer by the fund's manager to Dr. Bruno would be that ratchet and DPI, respectively, is:

	Ratchet	DPI	
A)	The year the fund was set up	Dividends paid out as a fraction of paid-in capital	✗
B)	The allocation of equity between shareholders and management	The limited partner's realized return from the fund	✓
C)	The general partner's share of fund profits	The general partner's realized return	✗

Explanation

Ratchet is a contract term that specifies the allocation of equity between management and shareholders.

DPI, or distributed to paid-in capital, is the cumulative distributions paid out from the fund as a fraction of cumulative invested capital. DPI measures the limited partners' realized return from the fund.

Note: The GP's share of fund profits is called *carried interest*. The year the fund was set up is called the *vintage*. There should be no distinction between realized and unrealized return for the GP. Also, there is no term for dividends over paid-in capital as dividends are seldom paid out from a private equity fund.

(Module 33.3, LOS 33.i)

Question #58 of 59

Question ID: 1473844

A private equity investor makes a $5 million investment in a venture capital firm today. The investor expects to sell the firm in four years. He believes there are three equally possible scenarios at termination:

1. expected earnings will be $20 million, and the expected P/E will be 10.
2. expected earnings will be $7 million, and the expected P/E will be 6.
3. expected earnings will be zero if the firm fails.

The investor believes that ROI of 2.44x is appropriate. The expected terminal value and the investor's pre-money valuation, respectively, are *closest* to (in $ million):

	Expected terminal value	Pre-money valuation
A)	$121.00	$56.39
B)	$80.67	$28.06
C)	$9.00	$3.69

Explanation

The terminal value under each scenario is the expected earnings multiplied by the P/E ratio. The expected terminal value is the weighted average of the three scenarios (all in $ million):

Scenario 1: Terminal value = $20 × 10 = $200

Scenario 2: Terminal value = $7 × 6 = $42

Scenario 3: terminal value = $0

Expected terminal value = ($200 + $42 + $0) / 3 = $80.67

The expected terminal value is then divided by the ROI of 2.44 to arrive at the post-money (POST) valuation:

POST = terminal value / ROI = $80.67 / 2.44 = $33.06

The pre-money (PRE) valuation is the post-money valuation less the investor's initial investment:

PRE = POST − INV = $33.06 − $5.0 = $28.06

(Module 33.1, LOS 33.d)

Question #59 of 59

Question ID: 1473870

Which of the following terms correctly describes the risk to a private equity firm in long-term interest and exchange rates, and the provision that specifies the method of profit distribution between the limited partners (LPs) and general partner (GP), respectively?

	Risk in long-term rates	Profit distribution
A)	Market risk	Carried interest
B)	Market risk	Distribution waterfall
C)	Capital risk	Carried interest

Explanation

Market risk describes the risk of how changes in interest rate, exchange rate and other macroeconomic factors affect private equity investments.

The method of profit distribution between the LPs and GP is called distribution waterfall.

Carried interest is the GP's share of fund profits. Capital risk refers to the risk of capital depletion in a private equity fund and the risk of obtaining additional financing.

(Module 33.2, LOS 33.f)